

For my ClimaAware project I used the Arduino Framework in C++ on an ESP32.
framework: I used meta Ai to help me to finish writing my code .

Design Tool: I used Wokwi Simulator to design and test my ClimaAware system before building hardware.

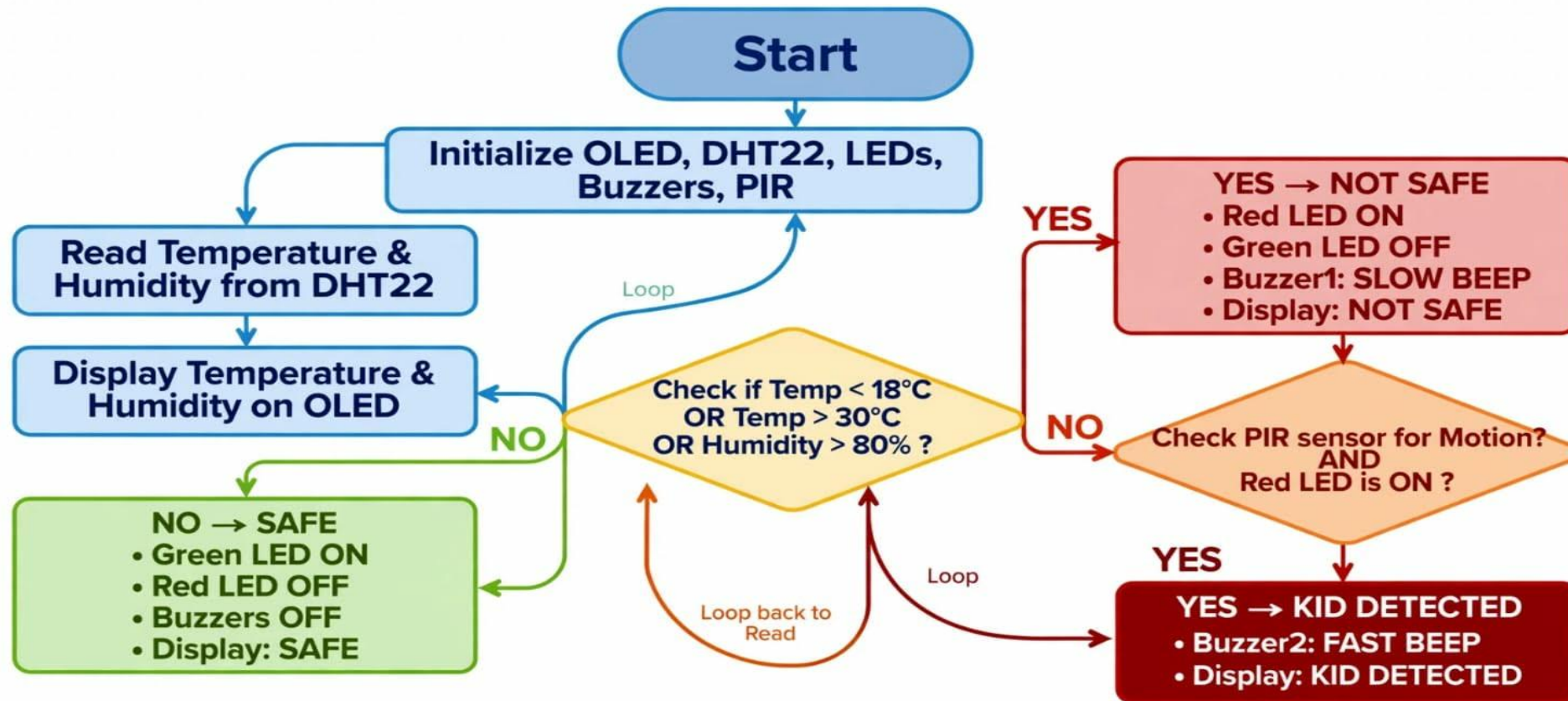
Programming Methods:

I implemented a threshold-based control system. The method checks if Temperature is less than 18°C or greater than 30°C or Humidity greater than 80%. If yes, the system enters NOT SAFE state. It also includes motion-based intervention where if PIR detects a child while in NOT SAFE state, it activates a fast buzzer. Frameworks: Language is C++ in Arduino IDE 2.x. Framework is Arduino ESP32. Libraries used are Adafruit_SSD1306 and GFX for OLED display and DHT sensor library for DHT22. Designs: The software design is a Finite State Machine (FSM) combined with modular functions: readSensors(), updateDisplay(), checkSafety(), and alertKid(). The design uses a continuous loop (polling architecture). Flowchart: Attached flowchart illustrates the complete program flow from Start, Initialize, Read Sensors, Display, Check thresholds, LED and Buzzer control, PIR check, Loop back.

FLOWCHART

ClimaAware Child Safety System

Arduino Project — Program Flowchart



Continuous Loop • The system continuously repeats to monitor temperature, humidity, and motion for child safety